

SUPERSTORE SALES DATA ANALYSIS

Submitted By:

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Tools & Technologies:

Python | Pandas | NumPy | Matplotlib | Jupyter Notebook

Project Objective

To analyze Superstore sales data and identify meaningful patterns in sales, products, customers, regions, customer segments, and order trends, and to provide data-driven business recommendations.

Acknowledgement

I would like to express my sincere gratitude to my internship mentors and everyone who supported me during the completion of this Data Analytics project.

This internship provided me with an opportunity to apply my knowledge of Python, data analysis, data visualization, and business analytics to a practical dataset. Working on this project helped me improve my analytical thinking and gain hands-on experience with tools such as Pandas, NumPy, Matplotlib, and Jupyter Notebook.

I am thankful for the guidance and learning opportunities provided throughout the project, which helped me complete this work successfully.

Apoorva Rai

Introduction

Data analytics helps organizations convert raw data into useful information for business decision-making. Sales datasets contain valuable information about customers, products, regions, orders, and revenue.

This project focuses on analyzing a Superstore sales dataset using Python. The analysis examines sales performance from different business perspectives, including geographical regions, product categories, sub-categories, individual products, customer segments, customers, and time periods.

The project uses exploratory data analysis and visualization techniques to identify important trends and patterns.

The final objective is to transform the analyzed data into meaningful business insights and practical recommendations.

Project Objectives

The major objectives of this project are:

- To understand the structure of the Superstore dataset.
 - To perform basic data quality checks.
 - To calculate important sales KPIs.
 - To analyze sales performance across regions.
 - To compare product categories and sub-categories.
 - To identify yearly and monthly sales trends.
 - To identify top and bottom-performing products.
 - To analyze customer segments.
 - To identify high-value customers.
 - To analyze order trends.
 - To derive meaningful business insights.
 - To provide business recommendations.
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Dataset Description

The dataset contains transactional information related to orders, customers, products, locations, shipping, and sales.

Important columns include:

Column	Description
Order_ID	Unique order identifier
Order_Date	Date of order
Ship_Date	Date of shipment
Customer_ID	Customer identifier
Customer_Name	Customer name
Segment	Customer segment
Region	Geographic region
Category	Product category
Sub_Category	Product sub-category
Product_ID	Product identifier
Product_Name	Product name
Sales	Sales amount
Order_Year	Year of order
Order_Month	Month of order

Dataset Limitation

The dataset does **not contain a Profit column**, so profitability analysis was not performed.

row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product
1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-10001
2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-10001
3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-10000
4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-10000
5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-10000

Tools and Technologies

Python

Used as the primary programming language for data analysis.

Pandas

Used for data loading, cleaning, grouping, aggregation and analysis.

NumPy

Used for numerical calculations.

Matplotlib

Used for creating charts and visualizations.

Jupyter Notebook

Used as the main environment for performing and documenting the analysis.

GitHub

Used to store and present the completed project.

METHODOLOGY

The project followed a structured data analytics workflow.

Step 1 — Data Loading

The dataset was loaded into a Pandas DataFrame.

Step 2 — Data Inspection

The dataset structure, columns, data types, and sample records were examined.

Step 3 — Data Quality Checking

The dataset was checked for potential data quality issues such as missing values and inconsistent information.

Step 4 — KPI Analysis

Important business-level metrics such as total sales and number of orders were calculated.

Step 5 — Exploratory Data Analysis

Sales were analyzed across:

- Regions
- Categories
- Sub-categories
- Products
- Customers
- Customer segments
- Years
- Months

Key Performance Indicators

The analysis calculated important overall business metrics.

Total Sales

\$2,261,536.78

Total Unique Orders 4,922

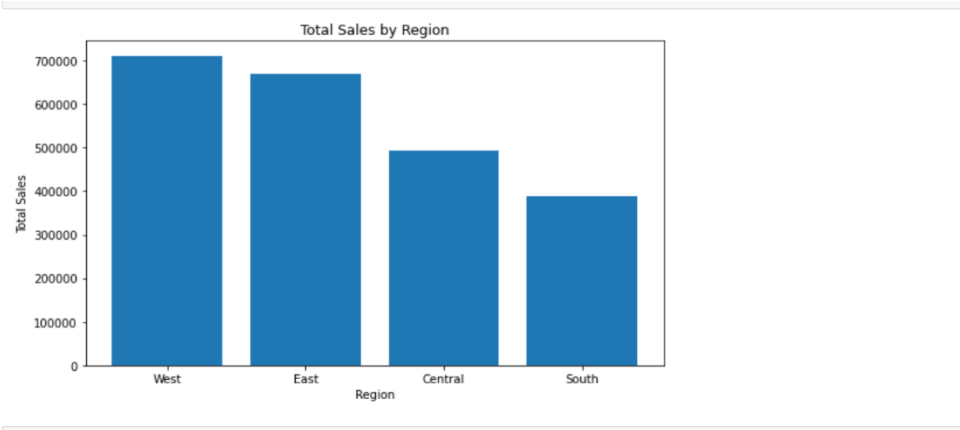
These KPIs provide a high-level understanding of the sales activity represented by the dataset.

Regional Analysis

Sales were analyzed across different geographic regions to identify regional differences in business performance.

Regional analysis helps understand which markets contribute strongly to overall sales and where region-specific strategies may be useful.

Sales by Region



Insight:

The regional analysis shows that sales performance differs across geographical regions. Understanding these differences can help businesses develop region-specific sales and marketing strategies.

Category Analysis

Sales were grouped by product category to compare the contribution of different major product groups.

Category-level analysis provides a broad understanding of product performance.

Sales by Category



Insight

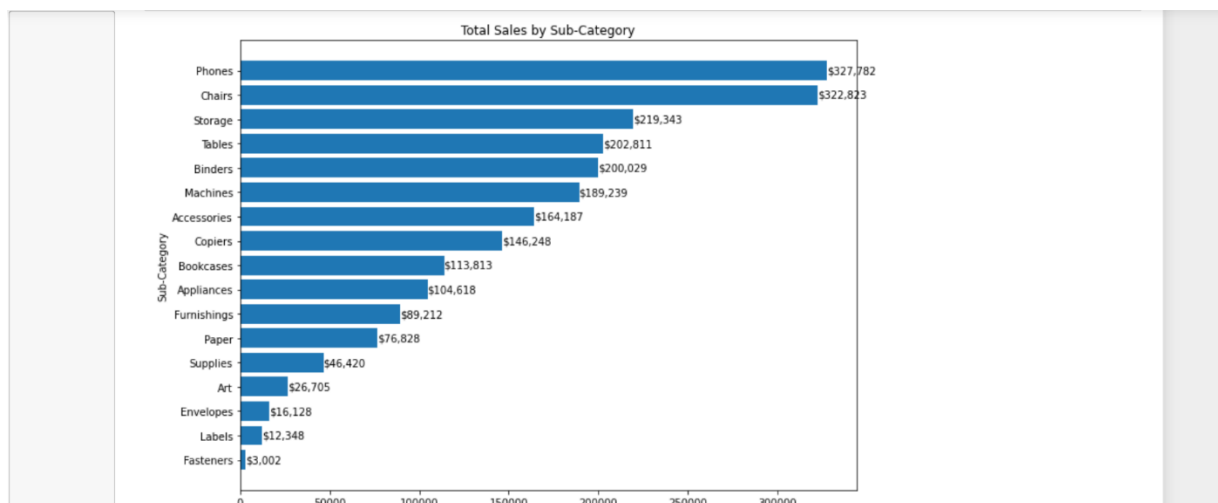
The category comparison helps identify which major product groups contribute most to overall sales and provides a basis for more detailed sub-category analysis.

Sub-Category Analysis

Sub-category analysis provides a more detailed view of product performance.

Sales were grouped and compared across individual sub-categories.

Sales by Sub-Category



Insight

The analysis demonstrates that sales performance varies across sub-categories. High-performing sub-categories can receive greater attention, while weaker-performing areas can be investigated further.

TIME ANALYSIS

Yearly Sales Analysis

yearly sales and order activity were analyzed to identify long-term trends.

Year	Sales	Orders
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2015	\$479,856.21	947
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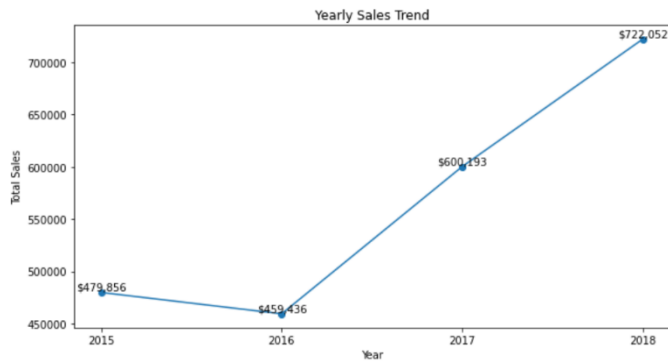
2016	\$459,436.01	1,019
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2017	\$600,192.55	1,295
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2018	\$722,052.02	1,661
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2018 was the strongest year, recording the highest sales and number of orders.

Yearly Sales Trend



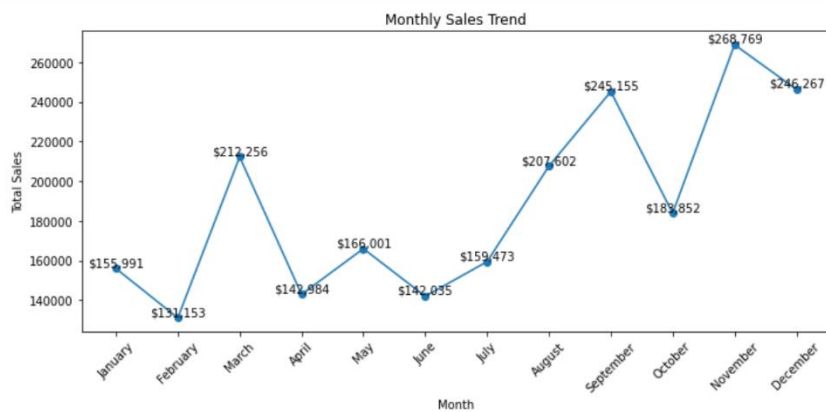
Insight

sales increased substantially from 2016 onward, reaching approximately \$722,052 in 2018.

Monthly Sales Analysis

Monthly sales were analyzed to identify variations in sales activity throughout the year.

Monthly Sales Trend



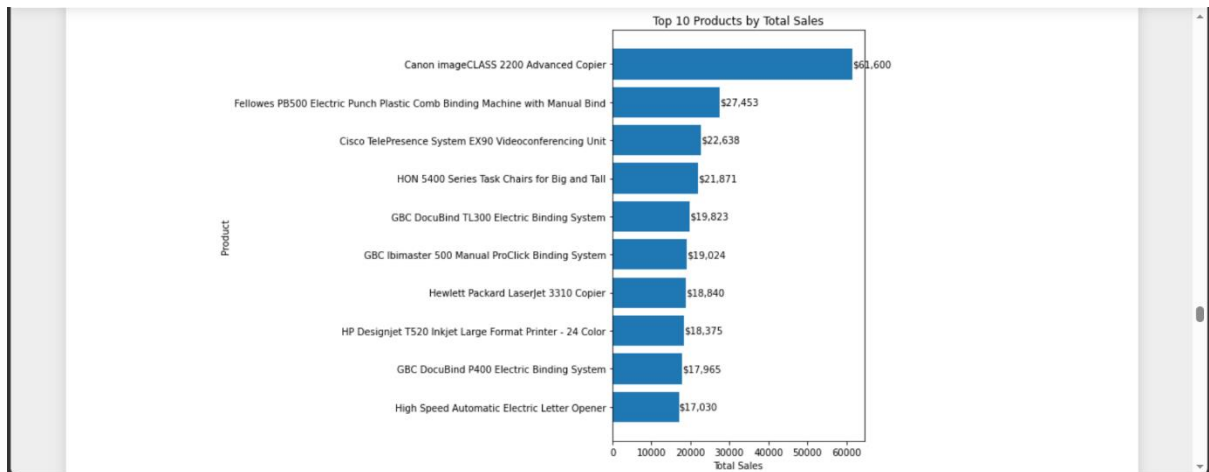
Insight

Monthly analysis helps identify periods of higher and lower sales activity and can support future inventory, marketing and sales planning.

Top 10 Products

The top 10 products were identified by calculating total sales for each individual product.

Top 10 Products by Sales



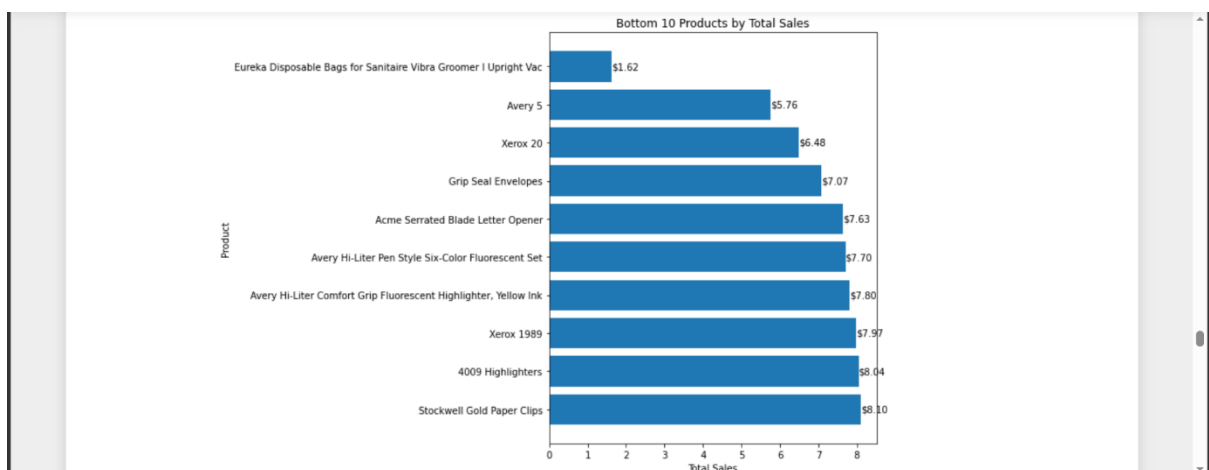
Insight

Top-performing products represent important contributors to overall sales and may deserve greater attention in inventory management and promotional activities.

Bottom 10 Products

The bottom 10 products were also identified based on total sales.

Bottom 10 Products by Sales



Important Insight

The lowest-selling product recorded approximately \$1.62 in total sales.

However, low sales alone should not be used to decide whether a product should be discontinued. Additional factors such as demand, pricing, inventory and profitability should be considered.

Customer Segment Analysis

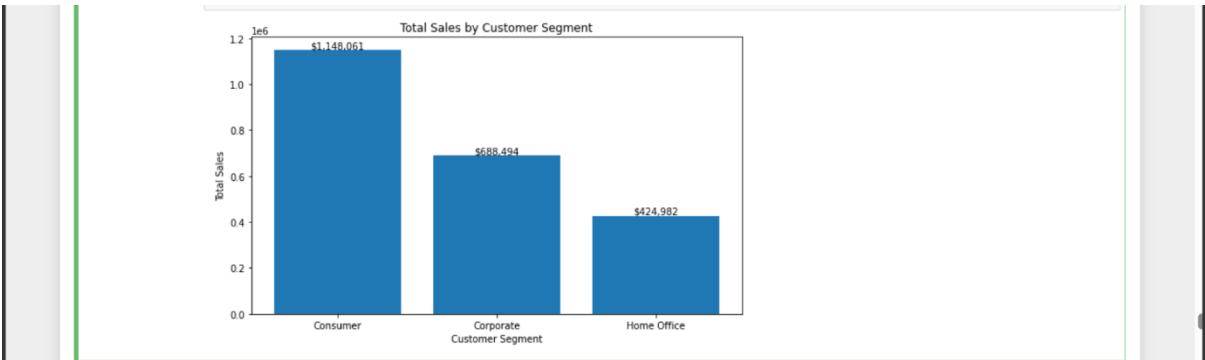
The dataset contains three customer segments:

- Consumer
- Corporate
- Home Office

Results

Segment	Total Sales	Contribution
Consumer	\$1,148,060.53	50.76%
Corporate	\$688,494.07	30.44%
Home Office	\$424,982.18	18.79%

Sales by Customer Segment



Key Insight

The Consumer segment generated the highest sales, contributing approximately 50.76% of total sales.

Corporate and Home Office segments represent additional opportunities for growth.

Customer Analysis

Customer-level analysis was performed to identify the highest-selling customers.

Top Customers

Customer	Total Sales
Sean Miller	\$25,043.05
Tamara Chand	\$19,052.22
Raymond Buch	\$15,117.34
Tom Ashbrook	\$14,595.62
Adrian Barton	\$14,473.57

Key Insight

Sean Miller was the highest-selling customer, generating approximately **\$25,043.05** in total sales.

High-value customers can be important targets for customer retention strategies.

Regional Sub-Category Analysis

The analysis compared sub-category performance across regions.

Region	Top Sub-Category	Sales
West	Chairs	\$100,023.20
East	Phones	\$99,884.66
Central	Chairs	\$82,372.78
South	Phones	\$58,098.34

Insight

Product demand varies across regions.

Chairs performed strongly in the West and Central regions, while Phones performed strongly in the East and South.

This suggests that region-specific product strategies may be useful.

Sales and Order Analysis

The dataset contains:

4,922 Unique Orders

The number of orders changed as follows:

Year Orders

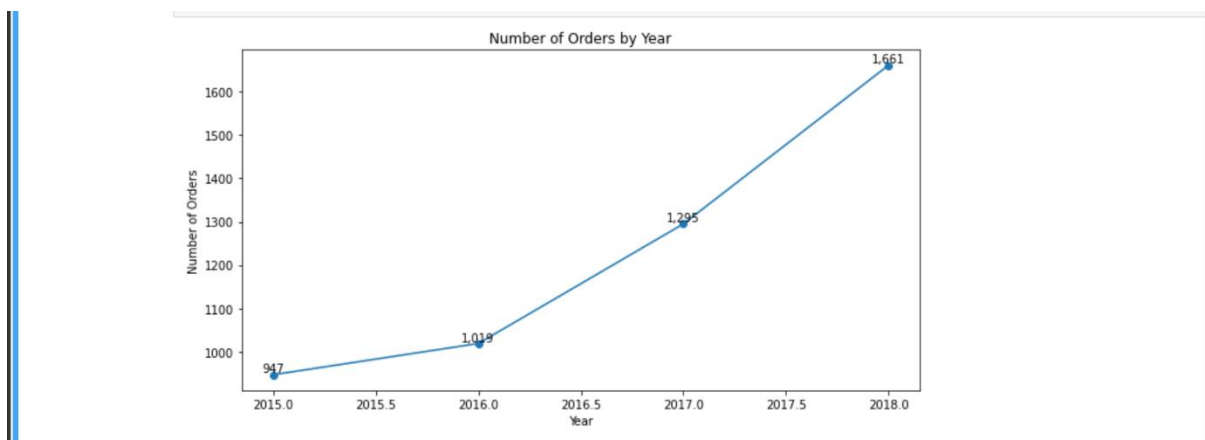
2015 947

2016 1,019

2017 1,295

2018 1,661

Number of Orders by Year



Insight

Order volume increased from **947 orders in 2015** to **1,661 orders in 2018**, showing strong growth in order activity.

An interesting observation is that 2016 had more orders than 2015 but lower total sales, showing that order count alone does not fully explain sales performance.

Final Business Insights

The complete analysis produced the following major findings:

1. Consumer Segment Dominance

Consumer customers contributed approximately **50.76% of total sales**, making them the largest segment.

2. Strong Sales Growth

Sales increased from approximately **\$479,856 in 2015** to **\$722,052 in 2018**.

3. Strongest Year

2018 was the strongest year, with approximately **\$722,052.02 in sales** and **1,661 orders**.

4. Growing Order Activity

Unique orders increased from **947 in 2015** to **1,661 in 2018**.

5. Regional Differences

Different regions showed different leading product sub-categories.

6. High-Value Customers

Sean Miller was the highest-selling customer with approximately **\$25,043.05** in sales.

7. Product-Level Variation

The analysis showed substantial differences between the sales performance of individual products.

RECOMMENDATIONS & CONCLUSION

Business Recommendations

Based on the analysis, the following recommendations are proposed:

Consumer Segment

Continue strengthening the Consumer segment because it contributes more than half of total sales.

Corporate & Home Office

Develop targeted strategies to increase sales from these segments.

Regional Strategy

Use regional sales patterns to develop location-specific product promotions.

High-Value Customers

Focus on retaining high-value customers through personalized offers and customer engagement.

Low-Performing Products

Investigate low-performing products before making inventory or discontinuation decisions.

Sales Growth

Study the strategies that contributed to strong growth from 2016 to 2018.

Order Monitoring

Regularly monitor order volume and average sales per order to understand purchasing behavior.

Limitations

The project has the following limitations:

- The dataset does not contain a Profit column.
 - Profitability analysis could therefore not be performed.
 - Information such as product cost, inventory and marketing expenditure was unavailable.
 - Low sales do not necessarily indicate low profitability.
 - The analysis is based on the historical data available in the dataset.
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Conclusion

The Superstore Sales Data Analysis project successfully demonstrated how Python-based data analytics can be used to understand business performance.

The analysis covered sales, regions, categories, sub-categories, products, customers, customer segments, and order trends.

The Consumer segment was the largest contributor to sales, while 2018 recorded the strongest overall performance.

The analysis also demonstrated that product demand varies across regions and that high-value customers can represent important opportunities for customer retention.

Overall, the project transformed raw transactional data into meaningful business insights and recommendations using Python, Pandas, NumPy, Matplotlib and Jupyter Notebook.

Future Scope

The project can be extended in the future by:

- Adding profit data for profitability analysis.
- Building sales forecasting models.
- Creating an interactive Power BI dashboard.

- Performing customer segmentation.
- Analyzing discounts and product quantity.
- Studying average order value.
- Developing predictive sales models.
- Performing advanced regional forecasting.

THANK YOU